# MACROECONOMICS

MACRO-38

**Time:** 5 minutes

**Outcome:** Apply the natural-rate hypothesis and vertical long-run Phillips curve.

**Difficulty:** Intermediate

# Long-Run Phillips Curve

## THE CORE IDEA

The long-run Phillips curve is vertical at the natural unemployment rate because expected inflation eventually adjusts. A change in labor-market structure can shift the vertical LRPC to a new natural rate. Expected inflation adjusts, the SRPC shifts, and unemployment returns toward its natural rate. The long-run Phillips curve is vertical at the natural rate of unemployment.

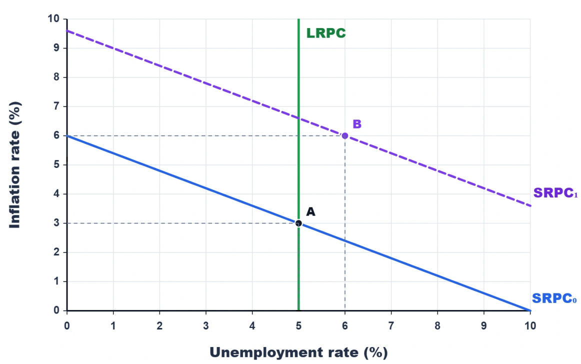
## HOW TO RECOGNIZE IT

- The long-run Phillips curve is vertical at the natural rate of unemployment.
- Expected inflation rises from 2 percent to 5 percent while unemployment remains at its natural rate.
- The LRPC is vertical at the natural rate because expected inflation adjusts in the long run.
- A national job-matching system reduces the time workers spend searching for suitable positions.

## WATCH OUT

Do not treat a higher expected inflation rate as a permanent way to lower unemployment. Along the long-run Phillips curve, different inflation rates can coexist with the same natural rate of unemployment.

## WORKED EXAMPLE: RETURNING TO THE NATURAL RATE




The vertical LRPC is drawn at 5% unemployment. A temporary demand expansion can move unemployment below 5% along a short-run Phillips curve, but expected inflation eventually adjusts and the short-run curve shifts. Unemployment returns to the 5% natural rate at a higher inflation rate, so no permanent tradeoff remains.

## CHECK YOURSELF

Why does higher inflation not permanently lower unemployment in the natural-rate model?

**READY?** Return to the game and master this concept.